

- 4 (a) (i) The electric field strength due to each particle is directed towards the left. As the resultant electric field strength at any point is the vector addition of the individual electric field strengths of A and B, it is always towards the left. Hence there will be no point where the electric field strength is zero.
- (ii) The electric potential due to A is negative while that due to B is positive. The total electric potential at any point is the scalar addition of the individual electric potentials of A and B. Hence, there will be a point in between the charges where electric potential is zero.
- (b) (i) By the principle of conservation of energy, considering energy changes of particle from the point it enters the electric field to the point it hits point P,

increase in kinetic energy = decrease in electric potential energy

$$\frac{1}{2}m(v_f^2 - v_i^2) = q\left(\frac{1}{2}\Delta V\right)$$

$$\begin{aligned}\Delta V &= \frac{m}{q}(v_f^2 - v_i^2) \\ &= \frac{(6.6 \times 10^{-27})}{(3.2 \times 10^{-19})} \left[(6.5 \times 10^5)^2 - (4.1 \times 10^5)^2 \right] \\ &= 5247 = 5250 \text{ V}\end{aligned}$$

Since the negatively charged particle accelerates towards plate Y which is at 0 V, plate X must be at a lower potential with respect to plate Y.

$$V = -5250 \text{ V}$$

(ii) $F_E = ma$

$$q\left(\frac{\Delta V}{d}\right) = ma$$

$$\begin{aligned}a &= \frac{q\Delta V}{md} \\ &= \frac{(3.2 \times 10^{-19})(5250)}{(6.6 \times 10^{-27})(3.6 \times 10^{-2})} \\ &= 7.0707 \times 10^{12} = 7.07 \times 10^{12} \text{ ms}^{-2}\end{aligned}$$

- (iii) Take direction to the right and downwards as positive.
consider horizontal motion,

$$s = -(u \sin \theta)t + \frac{1}{2}at^2$$

$$0.5 \times 3.6 \times 10^{-2} = -(4.1 \times 10^5 \sin 32^\circ)t + \frac{1}{2}(7.07 \times 10^{12})t^2$$

$$t = 1.0842 \times 10^{-7} \text{ s} \quad \text{or} \quad -4.6963 \times 10^{-8} \text{ s (NA)}$$

consider vertical motion,

$$d = (u \cos \theta)t$$

$$\begin{aligned}&= (4.1 \times 10^5 \cos 32^\circ)(1.0842 \times 10^{-7}) \\ &= 0.037698 = 0.0377 \text{ m}\end{aligned}$$

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5. (a) The sections of the disc moving towards or away from the observer experience a change